

Gauge Theory in $\mathcal{UD}$

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Abstract

This notebook extends the Relativity Toolkit (v1.11.0) from general relativity into basic Abelian scalar gauge theory. We demystify “gauge field” by relating it to the familiar concepts of coordinate system and reference frame. We find that *all forces arise from choice of gauge*, specifically from the implied connections. Inertial forces, like centrifugal and Coriolis forces, arise from choosing a rotating reference frame. Fundamental forces, like electromagnetic and nuclear forces, arise from choosing internal phase “reference-frame origins.” The difference between inertial forces and fundamental forces is that no choice of gauge (coordinate system or reference frame) can make fundamental forces go away. They may change in magnitude or other properties, depending on gauge choices, but never vanish completely or forever. Gravitational tidal forces from geodesic deviation are also fundamental in this way: no choice of coordinate system or reference frame can make them go away. In all cases, invariance of the physics, aka gauge symmetries, imply conservation laws.

In the $\mathcal{UD}$ (up-down) tensorial calculus, we implement a new, generalized covariant-derivative operator that handles both the spacetime geometry of general relativity and the internal phase geometry of electrodynamic and weak gauge connections, simultaneously. We demonstrate this new covariant derivative operator through four automated, symbolic proofs:

- *Metric Compatibility*: the Levi-Civita connection uniquely nulls the gradient of the metric ($\nabla g = 0$)
- *$U(1)$ Gauge Covariance*: the covariant derivative of a scalar field transforms via local phase rotation
- *Field Strength Invariance*: the electromagnetic tensor $F_{\mu\nu}$ is invariant under local gauge transformations
- *Charged Vector Boson Covariance*: In the $U(1)$ sector of Weinberg-Salam electroweak theory, we show covariance for a vector field W^α that enjoys both spacetime and internal phase invariance

Finally, we apply the updated engine to re-derive the Riemann curvature tensor from the commutator of covariant derivatives.

- A .wl script version of code for this notebook is available at
“https://raw.githubusercontent.com/rebcabin/RelativityToolkit/main/non_quantum_gauge_theory.wl”

Initialize the Relativity Toolkit

Optional: Quit for a Fresh Kernel

- *Note, if you call `Quit` in a notebook, evaluation may hang. If you want to “Evaluate Notebook,” mark the following two cells “Evaluatable” in the Cell→Property menu, call `Quit` and `FrontEndTokenExecute` once each, then comment out the cells, or mark them unevaluatable again. Alternatively, you can quit the kernel via the Evaluation menu (last item). At that point, you can “Evaluate Notebook” in a clean environment, or evaluate cells individually.*

```
Quit[];
```

```
In[*]:= FrontEndTokenExecute["DeleteGeneratedCells"]
```

Download the Code

```
In[*]:= RTbranch = "v1.11.0";
RTbase = "https://raw.githubusercontent.com/rebcabin/RelativityToolkit/" <>
  RTbranch <> "/";
RTbust = "?t=" <> ToString[Round[AbsoluteTime[]]]; (*Force fresh download*)
RTrules = RTbase <> "RelativityToolkit.wl" <> RTbust;
RTtests = RTbase <> "RelativityToolkit_RegressionTests.wl" <> RTbust;
RTviz = RTbase <> "RelativityToolkit_VisualShowcase.wl" <> RTbust;
Get[RTrules];
```

» Relativity Toolkit version : 1.11.0

» Relativity Connection Symbol : Γ

Optional: See all Tests and Results from Prior Works

Make any of the following cells evaluatable if you wish. They’re unevaluatable by default to keep this notebook trim.

```
In[*]:= Get[RTtests];
In[*]:= Get[RTviz] (* no semicolon, on purpose *)
In[*]:= Get[RTbase <> "covariant_derivative.wl"]
In[*]:= Get[RTbase <> "law_of_gamma.wl"]
In[*]:= Get[RTbase <> "connections_to_christoffels.wl"]
Get[RTbase <> "schwarzschild_from_scratch.wl"]
```

All Physics is Gauge Theory

Look at the dictionary definition of “gauge:”

1. *n.* A device or instrument for measuring the magnitude, amount, or contents of something. Ex.: a fuel gauge
2. *n.* The thickness, size, or capacity of something, especially as a standard. Ex.: the gauge of a rail line.
3. *v.* To estimate or determine the magnitude, amount, or volume of something. Ex. “Gauge the intrinsic luminosity of a star.”
4. *v.* To measure the dimensions of an object with a gauge instrument. Ex. “Gauge the assemblies and plane them to width.”

All this is concrete, referring to ordinary objects and activities in the trades or in the physics lab. None of it hints at the abstractions of gauge theory for electrodynamics, the nuclear forces, gravitation, let alone quantum field theory. Where are covariance, invariance, symmetry groups, conservation, unitarity, fibre bundles, and so on, in these definitions?

The answer is the other way around: these abstractions are not definitional, they’re mathematical machinery for various kinds of gauging, in the sense of measuring. Physics is, fundamentally, the science of measurement, and we can’t measure anything without setting up gauges. Most familiarly, we choose coordinate bases and origins for measuring positions and momenta in spacetime. *Setting a gauge* is setting coordinate bases and origins for measurements. Likewise, we must choose a reference axis for measuring the phase of an electromagnetic field. *Setting a gauge* for electrodynamics is setting such a phase reference at each point of spacetime. By analogy to position and momentum, we can say that setting the phase reference is *setting a coordinate system basis and origin for phase*.

It seems that coordinates, reference frames, bases, and origins are all so deeply related, regarding purpose and necessity, that we can call all of them “gauges” and we can call the activities of setting them up as “setting gauge” or “choosing gauge.”

Now, to the surprising statement: **All Forces Arise from Choice of Gauge.**

We’re not accustomed to this idea. We imagine forces as intrinsic physics, independent of gauges set to measure them. Planets are attracted to the Sun by a gravitational force, and we get compatible results in all reference frames. Electrons float around atomic nuclei by the same attractive Coulomb force, no matter how we measure it. Springs and pulleys apply forces to loads, no matter how we measure them.

But forces change motions, and motions must be measured in coordinate systems, i.e., gauges. Different choices of gauge yield different values for motion measurements, so different gauges imply different forces to produce the observed motions. Some of these forces are only apparent, such as centrifugal and Coriolis forces. We can make these go away by choosing different a different gauge, say an inertial frame. Other forces are such that no choice of gauge can make them go away. The electric force is like that, and so is geodesic deviation, the tidal force in curved spacetime. Choice of gauge will lead to different measurements, but invariances due to unavoidable symmetries in reveal conserved quantities, conserved in all gauges.

- When choice of the zero of stopwatches doesn’t affect the physics, when time symmetry obtains, then energy is conserved.

- When choice of the zero of position measurements doesn't affect the physics, when position symmetry obtains, momentum is conserved.
- When choice of a zero for phase doesn't affect the physics, when phase symmetry obtains, electric charge is conserved.

And so on. For instance, when unitary transformations don't affect nuclear physics, hypercharge and color charge are conserved.

We Hear Too Late about Gauge

In elementary physics, we don't call choice of reference frame basis and origin a "gauge choice." Nor do we hear the exotic idea that gauge choice generates forces. But such is the case. For an elementary example, consider measuring the altitude of an Earth satellite flying in a stable, circular, free-fall orbit. If we measure altitude with respect to mean sea level, the satellite seems to have a constant altitude, Earth's oblateness aside, and we measure no forces on it. However, if we measure altitude with respect to the irregular surface of the Earth, the satellite's altitude varies madly up and down. To naive observers, unaware of this poor choice of gauge, the satellite appears to suffer inexplicable forces pushing it up and down. However, these apparent forces arise merely from the choice of gauge. Here, "gauge" has its ordinary, dictionary meaning, as an altimeter with arbitrary, locally variable, origin setting. An even more elementary example is a rotating reference frame. Measure positions and momenta of free particles in an inertial lab frame, see no forces. Measure from a rotating frame, see centrifugal and Coriolis forces, arising merely from the choice of gauge, of reference frame.

With gauge theory now pertaining even to classical mechanics, why don't physics students study "gauge" before classical electrodynamics, where it is an abstract, unobservable, *arbitrary* field whose *gradient* is added to the vector potential, another unobservable field. How much farther from reality can you get? There is no conceivable relationship of this idea to the dictionary meaning of "gauge."

By the time our students get to electrodynamics, it's too late to go back and review the forces of elementary mechanics as arising from gauge. Our students will carry multiple pieces of baggage, all the way through general relativity (GR), with "gauge" a nebulous, ephemeral concept at best, until it appears again as even more abstract fields of matrices of complex numbers in particle physics.

Our students could have packed all these ideas into one piece of baggage with a concrete view of gauge as a choice of reference frame basis and origin, varying from the concrete to the abstract, from the scalar to tensor, from the Reals to the Complex, but just a reference frame nonetheless. Then, knowing that **all forces, apparent or fundamental, do, in fact, arise from choice of gauge**, and knowing that gauge symmetries drive conservation laws, we can put the travel sticker "Noether" on our one bag, get "Noether" stamped in our passport visas, and have a less fretful journey through all of physics.

Gauge Forces as Connections

How, exactly, do forces arise from gauge? Let's see how gauge *connections*, old friends from the covariant derivative, are the ultimate source of force. Recall from earlier notebooks that in GR, the Christoffel

symbols are the connections that separate intrinsic curvature of spacetime, which no choice of gauge can eliminate, from extrinsic curvature caused by choice of coordinates. At a single point, we can always choose a local inertial frame, a gauge, that makes Christoffel symbols vanish, and acceleration, thus force, go away. This is Einstein’s “elevator” demonstration of the Principle of Equivalence. But the paths of two particles, initially parallel, will converge or diverge in a curved spacetime, and no choice of gauge can make these “tidal forces” go away.

We’ve studied covariant derivative and connections in this series of notebooks, specifically in GR. We’ve repeatedly read that we can and should recast things as gauge theory, but we’re not told how. Now is the time, with the clarity that gauge choice is just choice of reference frame basis and origin, though in a generalized sense. *Geometrical* frame invariance, requiring that physic quantities transform tensorially amongst choice of time and space reference frames, drives the “discovery” of covariant derivative and connection. Connection drives equations of motion, thus forces, say for Schwarzschild orbits. We’re now equipped to study the other forces of Nature on the same terms.

Back to our physics students. This time, in life v2.0, with lighter, Noether-tagged baggage, they encounter gauge again, *not* for the first time, in classical electrodynamics. They realize that the gradient $\partial_\mu \Lambda$ of any differentiable scalar field Λ , rightfully called a gauge field, can be added to the vector potential A^μ without altering the physics:

$$\bar{A}_\mu(x^\nu) = A_\mu(x^\nu) + \partial_\mu \Lambda(x^\nu) \quad (1)$$

Ok, Λ is a gauge field, but in what sense is Λ the choice of a reference frame? For that matter, the choice of an uncountable infinity of reference frames, one at each spacetime point x^ν ?

Well, electrodynamical gauge invariance is *also* presented as **phase invariance**, a factor of $e^{i\Lambda}$ (same Λ) multiplied into a charged matter field ψ without altering the physics. Spoiler: choose phase with respect to an arbitrary zero. This is the *primary* principle of electrodynamical gauge as a coordinate frame basis and origin. Gradient added to vector potential (Equation 1) is just the *mechanism*, the “induced transformation,” that upholds that principle. Here’s how that works:

Change a matter field ψ by rotating its phase (gauge) by an arbitrary (but smoothly varying) amount, $\Lambda(x^\nu)$, at every point x^ν , without changing the geometrical coordinates x^μ :

$$\bar{\psi}(x^\nu) = e^{i\Lambda(x^\nu)} \psi(x^\nu) \quad (2)$$

This is just like choosing a reference-frame origin for the phase of the matter fields, locally, at each point, without changing the spacetime geometry x^ν .

Take the ordinary partial derivative, ∂_μ , of this new field $\bar{\psi}$. The ordinary product rule of differentiation spits out a “garbage term” because Λ varies with position:

$$\begin{aligned} \partial_\mu \bar{\psi}(x^\nu) &= e^{i\Lambda(x^\nu)} (\partial_\mu \psi(x^\nu) + i (\partial_\mu \Lambda(x^\nu)) \psi(x^\nu)), \text{ or, abbreviating away the spacetime coordinates} \\ \partial_\mu \bar{\psi} &= e^{i\Lambda} (\partial_\mu \psi + i(\partial_\mu \Lambda) \psi) \end{aligned} \quad (3)$$

Sound familiar? This is exactly what happened in relativity. The “garbage terms” (Hessians) drove “discovery” of covariant derivative and connection coefficients. Here, likewise, the extra term $i(\partial_\mu \Lambda) \psi$ ruins the physics.

As before, we find that the ordinary derivative is not covariant: it doesn’t survive changes in phase

coordinates, arbitrary changes in choice of $\Lambda(x^\mu)$.

In GR, the gauge was spacetime geometry itself, so compensating for coordinate change required 4D index manipulation. With electrodynamics, compensating for phase changes is easier. The *electrodynamical connection* is a one-index object because the electrodynamical gauge field Λ is one-dimensional.

To clean up the electrodynamical gauge garbage, we need a correction term in our covariant derivative of the matter field, just as we did with geometry and GR:

$$D_\mu \equiv \partial_\mu - iA_\mu \quad (4)$$

Yes, the connection is the vector potential, hence it drives electrodynamical forces, just as the Christoffel (Levi-Civita) connection drives gravitational forces. Equation 1 specifies how the connection changes with change of phase gauge. Notice this connection does not transform like ψ with change of Λ , just as the Christoffel symbols don't transform like tensors with change of spacetime coordinates. Connections aren't tensorial.

Now, change the geometrical, spacetime coordinate system leaving the phase fixed. For the garbage terms to cancel out exactly, A_μ must transform as in Equation 1 (abbreviated):

$$\bar{A}_\mu = A_\mu + \partial_\mu \Lambda \quad (5)$$

We have the new covariant derivative

$$D_\mu \psi \equiv \partial_\mu \psi - iA_\mu \psi \quad (6)$$

and

$$\begin{aligned} D_\mu \bar{\psi} &= \partial_\mu \bar{\psi} - i\bar{A}_\mu \bar{\psi} = e^{i\Lambda} (\partial_\mu \psi + i(\partial_\mu \Lambda) \psi) - i\bar{A}_\mu \bar{\psi} \\ &= e^{i\Lambda} (\partial_\mu \psi + i(\partial_\mu \Lambda) \psi) - i(A_\mu + \partial_\mu \Lambda) (e^{i\Lambda} \psi) \\ &= e^{i\Lambda} \partial_\mu \psi + e^{i\Lambda} i(\partial_\mu \Lambda) \psi - e^{i\Lambda} iA_\mu \psi - e^{i\Lambda} i(\partial_\mu \Lambda) \psi \\ &= e^{i\Lambda} \partial_\mu \psi - e^{i\Lambda} iA_\mu \psi \\ &= e^{i\Lambda} (\partial_\mu \psi - iA_\mu \psi) \end{aligned} \quad (7)$$

Thus $(D_\mu \psi)$ transforms “covariantly” in gauge space by multiplying by $e^{i\Lambda}$, even though A_μ does not transform covariantly in this way.

It so happens that A_μ is, geometrically, a covariant covector, so it *does* transforms covariantly with change in coordinates x^ν .

The geometrical covariant derivative from GR transforms covariantly with geometrical coordinates. The gauge covariant derivative, D , transforms covariantly with phase, i.e., gauge coordinates.

Two covariant derivatives, one against geometrical coordinates, the other against phase. One idea of covariance.

- We now see why it's a good idea, after all, to call the covariant derivative covariant. Such honors the metaphor between “spacetime coordinates as spacetime gauge with its own covariant derivative” and “phase as gauge everywhere with its own covariant derivative,” despite the fact that in geometry and GR, the covariant derivative also pertains to contravariant indices, a fact we dismissed as “historical” in an earlier notebook.

We'll write some new functions for the Relativity Toolkit to automate the calculation above, after just one last setup section.

Covariant Derivatives, Revisited

In GR, we take covariant derivatives of tensors that inhabit the tangent and cotangent tensor spaces at each point of a curved manifold. The canonical example is the covariant derivative of an arbitrary contravariant vector A^μ (A not for not vector potential here! A for arbitrary!) with respect geometrical coordinates:

```
In[ ] := With[{example = HoldForm[CD[A[U[μ]], x[U[ν]]]}],
  Echo[TensorForm@@example, example ""]];
```

$$\gg (A^\mu)_{;\nu} = A^\mu_{;\nu} + (A^\lambda) \Gamma^\mu_{\nu\lambda}$$

The connection coefficients Γ account for twisting and turning of basis vectors in the tangent spaces as the vector A^μ is parallel-transported geometrically. In GR, this twisting and turning causes geodesics to converge or diverge, thus causing tidal forces. Twisting and turning also drives single geodesics, giving rise to orbits, gravitational lensing, gravitational waves, and so on.

- *Emphasis: Geodesic deviation is indistinguishable from tidal force. It's not an apparent force: its numerical values may depend on gauge choice, but we can't get rid of it globally by gauge choice.*

The derivative above transforms like a tensor, ensuring physics is **locally Lorentz invariant**. This invariance invites Noether's theorem again, implying a conserved quantity. In GR, this conserved quantity is **spin current**, from symmetries of non-coordinate bases or **tetrads** (see Appendix 1).

“Wallpaper” Spaces Attached to Geometry

In this notebook, we consider new spaces “attached” to geometrical points of the manifold, like wallpaper. These spaces are gauge fields whose connections represent electromagnetism and nuclear forces. These wallpaper spaces have their own abstract differential geometries—their own covariant derivatives, above and beyond the differential geometry of spacetime itself. But there is a big difference:

In gravitation, gauge forces arise FROM the geometry. In other theories, gauge forces arise ON the geometry.

- *That's one reason gravitation resists unification with the other forces of Nature.*

While it is fine to analogize gauge fields as “wallpaper,” the rigorous mathematical machinery for this is **fibre bundles** (Appendix 2).

The objective of this notebook is to apply $\mathcal{UD}$ to these wallpaper gauge fields, encoding transformation laws, invariants, and conserved quantities with the same rigor we applied to GR.

A New CD

This new overload of CD can compute connections for Abelian (commutative) gauges like scalar phase.

These connections arise on **sections of the principal bundle** of Appendix 2. We'll get to connections for non-Abelian gauges like $SO(3)$ and $SU(2)$ in future notebooks.

Gauge Covariant Derivative

Inputs

1. **expr**: The field being differentiated, scalar or tensor
2. **var**: the coordinate function, e.g., $x[\mathcal{U}[\mu]]$, the differentiation variable
3. **q**: the coupling constant or “charge”
4. **A**: the gauge-field connection symbol. We keep Γ implicitly as the spacetime connection for all theories

```
In[*]:= CD[expr_, var_, q_, A_] :=
Module[{gravitationalPart, gaugeCorrection, idx},
(*Calculate the coordinate Covariant Derivative via existing CD overloads.*)
(*Handle  $\partial$  + Christoffels (if expr is a tensor)*)
(*If expr is a scalar, just return  $\partial$ .**)
gravitationalPart = CD[expr, var];
(*Extract the index from the differentiation variable.*)
(*var is some  $x[\mathcal{U}[\mu]]$ ; we need  $\mu$ *)
idx = var[[1, 1]];
(*Construct the Gauge Correction, e.g.,  $-i q A_\mu * \text{expr}$ *)
(*Note: This A has a DOWN (Covariant) index matching the derivative*)
gaugeCorrection = -I * q * A[D[idx]] * expr;
(*Combine*)
gravitationalPart + gaugeCorrection];
```

Electrodynamics Demos

Covariance of the Covariant Derivative

In “Forces as Gauge Connections,” we subjected the following expression to a 1D (Abelian) gauge transformation in a manual calculation. Let’s automate that calculation, now including a specific charge, q :

```
In[*]:= (cde = CD[ψ, x[ℳ[μ]], q, A])
Out[*]:= -i q ψ (A_μ) + ψ_{,μ}
```

Starting with the gauge transformations

$$\bar{\psi} = e^{iq\Lambda} \psi \quad \text{and} \quad \bar{A}_\mu = A_\mu + \partial_\mu \Lambda$$


```
In[8]:= gaugeTransforms = {
   $\psi \rightarrow e^{iq\Lambda} \psi,$ 
   $A[\mathcal{D}[\mu\_]] \mapsto A[\mathcal{D}[\mu]] + \text{Partials}[\Lambda, x[\mathcal{U}[\mu]]]$ 
}
```

```
Out[8]= { $\psi \rightarrow e^{iq\Lambda} \psi,$   $A_{\mu} \mapsto A_{\mu} + \Lambda_{,\mu}$ }
```

prove that the covariant derivative

$$D_{\mu} \psi \equiv \partial_{\mu} \psi - iq A_{\mu} \psi$$

transforms exactly the same way as ψ

$$D_{\mu} \bar{\psi} = e^{iq\Lambda} D_{\mu} \psi$$

We just need one rule for the charge constant (assumes, but does not prove, Noether!). Everything else is already in the toolkit's `differentiationRules`:

```
In[9]:= constantRules = {Partials[q, x[\mathcal{U}[\mu\_]]] \mapsto 0}
```

```
Out[9]= {q, \mu \mapsto 0}
```

```
In[10]:= transformedCde =
  (cde /. gaugeTransforms /. differentiationRules /. constantRules) // Expand //
  Simplify
```

```
Out[10]= e^{iq\Lambda} (-i q \psi (A_{\mu}) + \psi_{,\mu})
```

```
In[11]:= transformedCde === e^{iq\Lambda} cde
```

```
Out[11]= True
```

QED ■

Invariance of the Field-Strength Tensor

The field-strength tensor, $F_{\mu\nu}$, aka Faraday, is defined as follows

```
In[12]:= F[\mu, \nu] = Partials[A[\mathcal{D}[\nu]], x[\mathcal{U}[\mu]]] - Partials[A[\mathcal{D}[\mu]], x[\mathcal{U}[\nu]]]
```

```
Out[12]= - (A_{\mu, \nu}) + A_{\nu, \mu}
```

usually presented as $\partial_{\mu} A_{\nu} - \partial_{\nu} A_{\mu}$ to make the indices line up, but our comma notation checks out.

Now, subject it to the gauge transform:

```
In[13]:= F[\mu, \nu] /. gaugeTransforms
```

```
Out[13]= - (A_{\mu} + \Lambda_{,\mu})_{,\nu} + (A_{\nu} + \Lambda_{,\nu})_{,\mu}
```

See that it doesn't change by invoking the toolkit's `ExpandDerivatives`:

```
In[*]:= (F[μ, ν] /. gaugeTransforms) // ExpandDerivatives
Out[*]=

$$-\left(A_{\mu,\nu}\right) + A_{\nu,\mu}$$

QED ■
```

Charged Vector Boson Covariance Demo

In this demo, we differentiate a charged contravariant vector field W^α , coupled to a neutral field Z_μ by a scalar hypercharge g_W .

Z_μ acts like a vector phase correction in this phase covariant derivative. In addition to this vector phase correction, W^α generates a spacetime coordinate correction proportional to Γ , as with the spacetime covariant derivative of any tensor.

We did not need a geometrical correction in electrodynamics because we were differentiating a scalar field ψ . Its spacetime covariant derivative is exactly equal to its ordinary partial derivative—the twisting and turning of spacetime does not change a scalar value. Here, we do not need spacetime correction for Z_μ because Z_μ is not the subject of differentiation.

This expression occurs in the electroweak, $U(1)$ sector, an Abelian sector (multiple phase operations commute), of the Standard Model of Particle Physics.

We proceed just as above

```
In[*]:= (wsd = CD[W[U[α]], x[U[μ]], gW, Z]) // TensorForm
Out[*]=

$$W^{\alpha}_{,\mu} - i g_W (W^\alpha) (Z_\mu) + (W^\lambda) \Gamma^\alpha_{\mu\lambda}$$


In[*]:= mixedGaugeTransforms = {
  W[U[α_]] :-> e^{i g_W \Lambda} * W[U[α]],
  Z[D[μ_]] :-> Z[D[μ]] + Partials[Λ, x[U[μ]]]}
Out[*]=

$$\{W^\alpha \rightarrow e^{i g_W \Lambda} (W^\alpha), Z_{\mu} \rightarrow Z_{\mu} + \Lambda_{,\mu}\}$$


In[*]:= transformedWsd =
  (wsd /. mixedGaugeTransforms /. differentiationRules /. {Partials[gW, _] :-> 0}) //
  Expand // TensorForm
Out[*]=

$$e^{i g_W \Lambda} (W^{\alpha}_{,\mu}) - i e^{i g_W \Lambda} g_W (W^\alpha) (Z_\mu) + e^{i g_W \Lambda} (W^\lambda) \Gamma^\alpha_{\mu\lambda}$$


In[*]:= expectedWsd = wsd * e^{i g_W \Lambda} // Expand // TensorForm
Out[*]=

$$e^{i g_W \Lambda} (W^{\alpha}_{,\mu}) - i e^{i g_W \Lambda} g_W (W^\alpha) (Z_\mu) + e^{i g_W \Lambda} (W^\lambda) \Gamma^\alpha_{\mu\lambda}$$


In[*]:= transformedWsd === expectedWsd
Out[*]=
True
```

QED ■

Geometrical GR Demos

Geometrical Covariant Derivative of a Scalar Function Λ

equal to its ordinary partial derivative, rendered in comma notation

```
In[*]:= CD[Λ, x[U[μ]]]
Out[*]=
  Λ,μ
```

Geometrical Covariant Derivative of a Contravariant Vector V^μ

equal to its partial derivative plus the correction proportional to the geometrical connection Γ

```
In[*]:= CD[V[U[μ]], x[U[v]]] // TensorForm
Out[*]=
  V^μ,ν + (V^λ) Γ^μ_νλ
```

Geometrical Covariant Derivative of a (2, 0) Tensor

now we get two corrections, one for each index

```
In[*]:= CD[T[U[μ], U[v]], x[U[ρ]]] // TensorForm
Out[*]=
  T^μν,ρ + (T^λν) Γ^μ_ρλ + (T^μλ) Γ^ν_ρλ
```

Geometrical Covariant Derivative of a (1, 1) Tensor

ditto, with a sign change

```
In[*]:= CD[T[U[μ], D[v]], x[U[ρ]]] // TensorForm
Out[*]=
  T^μν,ρ - (T^μλ) Γ^λ_νρ + (T^λν) Γ^μ_ρλ
```

Geometrical Covariant Derivative of the Covariant Metric

Two negative signs, one for each covariant index

```
In[*]:= (cdg = CD[g[D[μ], D[v]], x[U[ρ]]]) // TensorForm
Out[*]=
  g_μν,ρ - (g_λν) Γ^λ_μρ - (g_μλ) Γ^λ_νρ
```

Metric Compatibility, $\nabla g = 0$

$\nabla g = 0$ was an axiom in earlier notebooks, and it yielded the Christoffel symbols in the *Fundamental Theorem of Riemannian Geometry*. Here, we do the opposite, showing that the Christoffel symbols

imply metric compatibility. Thus, metric compatibility is both necessary and sufficient for the Christoffel symbols.

- In abstract geometrical contexts, as opposed to coordinate-basis contexts, the Christoffel symbols are called the **Levi-Civita Connection**., often written $\left\{ \begin{smallmatrix} \lambda \\ \mu \nu \end{smallmatrix} \right\}$.
- Christoffel symbols were derived in Covariant Derivatives and Connections in the $\mathcal{UD}$ Calculus, assuming both torsion freedom (symmetry in lower two indices of Γ), and metric compatibility.
- If we allow torsion, as in Einstein-Cartan theory or as in Appendix 1, the present tautology is not valid. In such cases, we must introduce the **contortion tensor** K

$$\Gamma_{\mu\nu}^{\lambda} = \left\{ \begin{smallmatrix} \lambda \\ \mu \nu \end{smallmatrix} \right\} + K_{\mu\nu}^{\lambda} \quad (8)$$

We do not develop this case further in this notebook.

For pedagogy, clear the definitions loaded from the toolkit:

```
In[ ]:= ClearAll[ruleLeviCivita, cdgWithLeviCivitaConnection, metricDifferentiationRules];
```

Start with the covariant derivative of the metric, `cdg`, from above

Fetch the Levi-Civita connection via a new `ruleLeviCivita`, added to Relativity Toolkit v1.9.0:

$$\Gamma_{\mu\rho}^{\lambda} = \frac{1}{2} g^{\lambda\sigma} (g_{\sigma\rho,\mu} + g_{\sigma\mu,\rho} - g_{\mu\rho,\sigma}) \quad (9)$$

```
In[ ]:= ruleLeviCivita =  $\Gamma[\mathcal{U}[\lambda\_], \mathcal{D}[a\_], \mathcal{D}[b\_]] \Rightarrow$ 
Module[{ $\sigma = \text{Unique}["\sigma"]$ },
   $\frac{1}{2} g[\mathcal{U}[\lambda], \mathcal{U}[\sigma]] * ($ 
    Partials[g[ $\mathcal{D}[\sigma]$ ,  $\mathcal{D}[b]$ ], x[ $\mathcal{U}[a]$ ]] +
    Partials[g[ $\mathcal{D}[\sigma]$ ,  $\mathcal{D}[a]$ ], x[ $\mathcal{U}[b]$ ]] -
    Partials[g[ $\mathcal{D}[a]$ ,  $\mathcal{D}[b]$ ], x[ $\mathcal{U}[\sigma]$ ]] )]
```

```
Out[ ]:=
```

$$\Gamma_{a_b_}^{\lambda_} \Rightarrow \text{Module}[\{\sigma = \text{Unique}[\sigma]\}, \frac{1}{2} (g^{\lambda \sigma 23}) (g_{\sigma 23 b, a} + g_{\sigma 23 a, b} - g_{a b, \sigma 23})]$$

In `cdg`, replace Γ with its definition via this rule, then `Expand` to expose terms to further processing

```
In[ ]:= (cdgWithLeviCivitaConnection = (cdg /. ruleLeviCivita) // Expand) // TensorForm
```

```
Out[ ]:=
```

$$\begin{aligned} & -\frac{1}{2} (g^{\lambda \nu}) (g^{\lambda \kappa}) (g_{\kappa \mu, \rho}) - \frac{1}{2} (g_{\mu \lambda}) (g^{\lambda \kappa}) (g_{\kappa \nu, \rho}) - \\ & \frac{1}{2} (g^{\lambda \nu}) (g^{\lambda \kappa}) (g_{\kappa \rho, \mu}) - \frac{1}{2} (g_{\mu \lambda}) (g^{\lambda \kappa}) (g_{\kappa \rho, \nu}) + \\ & g_{\mu \nu, \rho} + \frac{1}{2} (g^{\lambda \nu}) (g^{\lambda \kappa}) (g_{\mu \rho, \kappa}) + \frac{1}{2} (g_{\mu \lambda}) (g^{\lambda \kappa}) (g_{\nu \rho, \kappa}) \end{aligned}$$

Now consider the following new `metricDifferentiationRules` in the toolkit:

They include

- metric symmetry: $g_{\mu\nu} = g_{\nu\mu}$, required by standard general relativity
- metric contraction to the Kronecker delta: $g_{\lambda\nu} g^{\lambda\mu} = \delta^\mu_\nu$ (gUpUp is inverse of gDownDown)
- contraction of Kronecker delta on derivatives, for the chain rule: $\delta^\mu_\nu \partial_\mu = \partial_\nu$
- delta contraction inside metric derivatives: $\delta^\mu_\nu g_{\mu\lambda,\rho} = g_{\nu\lambda,\rho}$, $\delta^\mu_\nu g_{\lambda\mu,\rho} = g_{\lambda\nu,\rho}$

```
In[*]:= metricDifferentiationRules = {
  (*Metric Symmetry*)
  g[D[a_], D[b_]] /; ! OrderedQ[{a, b}] => g[D[b], D[a]],
  g[U[a_], U[b_]] /; ! OrderedQ[{a, b}] => g[U[b], U[a]],
  (*Permissive Contraction (All 4 Alignments)*)
  (*First-First*)
  g[D[s_], D[a_]] * g[U[s_], U[b_]] => delta[U[b], D[a]],
  (*Second-Second*)
  g[D[a_], D[s_]] * g[U[b_], U[s_]] => delta[U[b], D[a]],
  (*First-Second (Redundant but Safe)*)
  g[D[s_], D[a_]] * g[U[b_], U[s_]] => delta[U[b], D[a]],
  (*Second-First*)
  g[D[a_], D[s_]] * g[U[s_], U[b_]] => delta[U[b], D[a]],
  (*Delta Contractions*)
  delta[U[bound_], D[free_]] * Partial[expr_, x[U[bound_]]] =>
    Partial[expr, x[U[free]]],
  delta[U[bound_], D[free_]] * Partial[g[D[bound_], D[other_]], v_] =>
    Partial[g[D[free], D[other]], v],
  delta[U[bound_], D[free_]] * Partial[g[D[other_], D[bound_]], v_] =>
    Partial[g[D[other], D[free]], v]};
```

The result $\nabla g = 0$ is immediate upon application of both the old metric rules and the new metric rules:

```
In[*]:= 0 ==
  (cdgWithLeviCivitaConnection //. (metricRules~Join~metricDifferentiationRules))

Out[*]=
  True

QED ■
```

Riemann Revisited

Here is a concise derivation of the Fundamental Theorem of Riemannian Geometry, shorter than the derivation in our earlier notebooks:

Calculate the commutator: $[\nabla_\rho, \nabla_\nu] A^\mu$:

```

In[*]:= term1 = CD[CD[A[U[μ]], x[U[ν]]], x[U[ρ]]];
term2 = CD[CD[A[U[μ]], x[U[ρ]]], x[U[ν]]];
(comm = term1 - term2) // TensorForm

Out[*]=

$$\left( \left( A^{\lambda} \right) \Gamma_{\nu\lambda}^{\mu} \right)_{,\rho} - \left( \left( A^{\lambda} \right) \Gamma_{\rho\lambda}^{\mu} \right)_{,\nu} - \left( A^{\mu}{}_{,\lambda} \right) \Gamma_{\nu\rho}^{\lambda} + \left( A^{\mu}{}_{,\lambda} \right) \Gamma_{\rho\nu}^{\lambda} - \left( A^{\lambda} \right) \Gamma_{\nu\rho}^{\kappa} \Gamma_{\kappa\lambda}^{\mu} +$$


$$\left( A^{\lambda} \right) \Gamma_{\rho\nu}^{\kappa} \Gamma_{\kappa\lambda}^{\mu} - \left( A^{\lambda} \right) \Gamma_{\rho\lambda}^{\kappa} \Gamma_{\nu\kappa}^{\mu} - \left( A^{\lambda}{}_{,\rho} \right) \Gamma_{\nu\lambda}^{\mu} + \left( A^{\lambda} \right) \Gamma_{\nu\lambda}^{\kappa} \Gamma_{\rho\kappa}^{\mu} + \left( A^{\lambda}{}_{,\nu} \right) \Gamma_{\rho\lambda}^{\mu}$$


Simplification Pipeline

In[*]:= calculatedRiemann =
  ((comm //
    ExpandDerivatives //
    Expand //
    CanonicalizeIndices) //.
    torsionRules) //
  ExtractCoefficient[#, A] & //
  TensorForm

Out[*]=

$$\Gamma_{\lambda\nu,\rho}^{\mu} - \Gamma_{\lambda\rho,\nu}^{\mu} - \Gamma_{\lambda\rho}^{\kappa} \Gamma_{\kappa\nu}^{\mu} + \Gamma_{\lambda\nu}^{\kappa} \Gamma_{\kappa\rho}^{\mu}$$


Textbook definition:  $\Gamma_{\nu\bar{s},\rho}^{\mu} - \Gamma_{\rho\bar{s},\nu}^{\mu} + \Gamma_{\rho\lambda}^{\mu} \Gamma_{\nu\bar{s}}^{\lambda} - \Gamma_{\nu\lambda}^{\mu} \Gamma_{\rho\bar{s}}^{\lambda}$ 

In[*]:= rawTextbook = Partials[Γ[U[μ], D[ν], D[s̄]], x[U[ρ]]] -
  Partials[Γ[U[μ], D[ρ], D[s̄]], x[U[ν]]] +
  Γ[U[μ], D[ρ], D[λ]] * Γ[U[λ], D[ν], D[s̄]] -
  Γ[U[μ], D[ν], D[λ]] * Γ[U[λ], D[ρ], D[s̄]]

Out[*]=

$$\Gamma_{\nu\bar{s},\rho}^{\mu} - \Gamma_{\rho\bar{s},\nu}^{\mu} - \Gamma_{\rho\bar{s}}^{\lambda} \Gamma_{\nu\lambda}^{\mu} + \Gamma_{\nu\bar{s}}^{\lambda} \Gamma_{\rho\lambda}^{\mu}$$


In[*]:= expectedRiemann = (rawTextbook //. torsionRules) // TensorForm

Out[*]=

$$\Gamma_{\lambda\nu,\rho}^{\mu} - \Gamma_{\lambda\rho,\nu}^{\mu} - \Gamma_{\lambda\rho}^{\kappa} \Gamma_{\kappa\nu}^{\mu} + \Gamma_{\lambda\nu}^{\kappa} \Gamma_{\kappa\rho}^{\mu}$$


In[*]:= calculatedRiemann === expectedRiemann

Out[*]=
True

QED ■

```

Appendix 1: Spin Connection as Gravitational Gauge

Gravitation acts *almost* like any other gauge theory, but not quite. Gravitational forces arise from spacetime connections **in the geometry**, while other force theories arise from non-coordinate connections **on the geometry**, in “wallpaper” spaces like $U(1)$ and $SU(3)$. This partial fit between gravitation and other gauge theories only deepens the frustration of unification-seekers.

To make gravitation a full gauge theory, we must find conserved quantities arising from local gauge

invariance. The definitive review paper is the following:

- Friedrich W. Hehl et al., *General Relativity with Spin and Torsion: Foundations and Prospects* (Rev. Mod. Phys. 48, 393, 1976)

which shows that:

- Invariance under general transformations (diffeomorphisms), implies conservation of the energy-momentum tensor.
- Invariance under local Lorentz transformations implies conservation of the spin angular momentum tensor (the spin current).
- Standard Einstein GR with vanishing torsion, which we assumed in this series of notebooks, collapses the two symmetries and hides the independent spin current.

Some other references are

- Sean Carroll, *Spacetime and Geometry: An Introduction to General Relativity*

- Appendix J, Page 486ff, under local Lorentz invariance, tetrads, and the spin connection:

Spin connection ω_{μ}^{ab} , arising from non-coordinate basis sets, acts as the gauge field for local Lorentz transformations. Carroll also explains that in standard GR the torsion tensor $T^{\lambda}_{\mu\nu}$ is zero, locking the Levi-Civita connection (Christoffel symbols) to the metric and forcing the energy-momentum tensor to be symmetric, absorbing spin effects via symmetrization.

- Also see Hehl above, Equation 3.8 and Sections III.B and III.C. In other references, this may be called *Belinfante-Rosenfeld symmetrization*.

- T. W. B. Kibble, *Lorentz Invariance and the Gravitational Field* (J. Math. Phys. 2, 212, 1961)

- *Foundational paper deriving gravitation as the gauge theory of the Poincaré group, i.e., translations + Lorentz*

Kibble shows that (1) gauging *translations* in Poincaré yields the tetrad field, defining the metric and conserving energy-momentum, and that (2) gauging the Lorentz *boost+rotation subgroup* yields the spin connection, permitting transformation of spinors, otherwise impossible.

- Video: Carmeci Academy. *General Relativity | Review and Intro to The Spin Connection*, <https://www.youtube.com/watch?v=LjydxYNY7Yg>

Appendix 2: Fibre Bundles

In this appendix, we introduce the standard mathematical language. We present *bundles*, *projections*, *fibres*, *sections*, *connections*, and (formally) *invariance*, with each concept applied to both geometry/-gravitation and to more general gauge fields, with electrodynamics the motivating example.

TL;DR: Dictionary

Out[] =

	Concept	Gauge Theory (EM)	Geometry / Gravitation	Interpretation
1	Principal Bundle	Phase Bundle (P)	Frame Bundle (LM)	All Possible Wallpapers
2	Principal Section	Local Phase Choice	Local Frame Choice (Tetrad)	Smooth Wallpaper Choice
3	Structure Group	U(1)	GL(4,ℝ) or SO(3,1)	Design Rules
4	Associated Bundle	Complex Line Bundle	Tangent Bundle (TM)	All Possible Walls
5	Associated Section	Charged Scalar Field	Vector Field V^μ	Particular Wall Choices
6	Connection	Vector Potential A_μ	Christoffel / Spin Connection	Wallpaper Glue

Bundles

Spaces attached to each point of a manifold are collected and flattened into sets called **bundles**. They are *fibre bundles* of projection functions (detail below).

In applications, there are two distinguished bundles: the **principal bundle** and the **associated bundle**, in a matched pair. Looking ahead, connections on sections of the principal bundle are “sources” of forces, and sections on associated bundles supply the quantities subject to forces, “victims” of forces. That’s a mnemonic slogan.

Geometry / Gravitation

Let $T_p M$ (an indivisible notation) be the set of all tangent vectors (with up indices) at point p of the manifold.

An **associated bundle** (victims of forces) of gravitation is the **tangent bundle** TM . This bundle is the **disjoint union** of all points p in the manifold paired with all vectors v ripped out of their tangent spaces at p , flattened into one big set of pairs:

$$TM \equiv \bigsqcup_{p \in M} \{v \in T_p M\} = \{(p, v) \mid p \in M, v \in T_p M\} \quad (\text{associated bundle, victims of forces}) \quad (10)$$

- Likewise, the cotangent bundle $T^* M$ is the set of all points paired with all covectors (down indices) at each point, and so on for each valence (combination of up-down indices) of tensor. These other geometrical bundles, for tensors of other valences, can also be associated bundles (victims of forces).

The **principal bundle** (sources of forces) of gravitation is the **frame bundle**, which supplies, at each point, all possible bases (gauges!) for frames of reference.

- All possible bases include the ordinary coordinate bases that conceal spin and the tetrad bases that reveal spin, sources of spin-coupled forces (Appendix 1).

Again, for emphasis, the **associated bundles** (victims of forces) are the associated tangent, cotangent, etc. bundles, which supply all possible tensors at each point.

Gauge Fields (EM)

With electrodynamics, imagine a copy of the symmetry group $U(1)$ (aka **structure group**) attached to each point $p \in M$. The members of $U(1)$ are phase factors, $e^{i\varphi}$ (imprecisely abbreviated as φ), each representing a choice of gauge at that point. The **principal bundle** (sources of forces) of electrodynamics is the disjoint union of all manifold points p paired with all possible phase factors at p , flattened into one big set of pairs.

$$U(1)M \equiv \bigsqcup_{p \in M} \{e^{i\varphi} \in U(1)\} = \{(p, \varphi) \mid p \in M, e^{i\varphi} \in U(1)\} \quad (\text{principal bundle, sources of forces}) \quad (11)$$

- We might call $U(1)M$ a phase bundle.

In classical (non-quantum) electrodynamics, the **associated bundle** (victims of forces) is the **complex line bundle**, a collection of pairs of a point and a copy of the entire complex plane at that point, all possible values, not necessarily of magnitude 1, of a complex-valued **charged matter field**. This field represents density and internal alignment (physical interpretation of phase of magnitude 1) of a notional charged fluid.

- *Emphasis: the copies of the complex plane in the associated bundle is the entire complex plane, not just the members of $U(1)$, which are confined to the unit circle in the complex plane.*

In quantum electrodynamics, the associated bundle is a collection of pairs of points and all possible Dirac wave functions, each of which is a superposition of spinor states. In this notebook, we cover only the classical case. The quantum case comes later in this series of notebooks.

Projections

Envision a **projection function**, π , that yields p from any element of TM :

$$\pi: TM \rightarrow M = (p, v) \mapsto p \quad (12)$$

“forgetting” the vector v .

In electromagnetism, the projection function $\pi: (U(1)M) \rightarrow M = (p, \varphi) \mapsto p$ forgets the phase factor $e^{i\varphi}$ from any given pair in $U(1)M$.

Fibres

In elementary mathematics, the **fibre**, $\text{fibre}(f, p) \equiv f^{-1}(p)$, of a function f at some point p is the preimage of f at p , that is, the subset of the domain of f that yield p .

- *In this usage, f^{-1} is not an “inverse function” that yields points, but a fibre-valued function that yields subsets. A general f may not be “invertible” in the usual sense of a point-to-point bijective function, but we can always define a fibre: a set containing all points of the domain that yield p .*
- *Unfortunately, we have the same notation, f^{-1} , both for a point-valued inverse function when f is invertible and for a set-valued, fibre-valued function whether f is invertible or not.*

The fibres of the bundles—principal or associated—of interest to us are the fibres of the projection functions π .

In geometry/gravitation, π^{-1} can't "remember" the forgotten vector, so it must return the set of all vectors attached to p (or, more generally, tensors):

$$\pi^{-1}(p) = \{v \in T_p M \mid \pi((p, v)) = p\} \equiv T_p M \quad (\text{victims of forces}) \quad (13)$$

which is, of course, the entire tangent space at p .

In electrodynamics, the π^{-1} can't remember the forgotten phase factor, so it must return a complete copy of $U(1)$ at each point:

$$\pi^{-1}(p) = \{\theta^{i\varphi} \in U(1) \mid \pi((p, \theta^{i\varphi})) = p\} \equiv U(1) \quad (\text{sources of forces}) \quad (14)$$

Sections

Geometry / Gravitation

A **section of the principal bundle** (sources of forces, the frame bundle) is a smooth choice of reference frame at each point—a coordinate basis for the Levi-Civita connection or a non-coordinate, tetrad basis for the spin connection.

A **section of the associated bundle** (victims of forces, the tangent bundle) is a smooth choice of one vector, $V^\mu(p)$, at each point, amounting to a smoothly varying vector field subject to forces. Likewise for other tensors.

Gauge Fields

In electrodynamics, a **section of the principal bundle** (sources of forces, the phase bundle) is a smooth choice of phase, $\varphi(p)$, at each point, which is a field of complex numbers of unit magnitude. It is the gauge-theory analog of a reference frame, a locally varying gauge choice.

A **section of the associated bundle** (victims of forces, matter fields) is a smooth choice of complex-valued charged matter field at each point (classical), or a smooth choice of Dirac wave function at each point (quantum).

Connections

Geometry / Gravitation

The connection on the gravitational principal bundle (sources of forces) is the field of Christoffel Symbols $\Gamma^\lambda_{\mu\nu}$, or the field of spin connections. These connections describe how an associated section—the vector field V^μ —twists and turns as p moves. A Levi-Civita connection is the Γ coefficient of the correction $V \cdot \Gamma$ that keeps the covariant derivative ∇V tensorial.

Gauge Fields

The connection on the principal gauge bundle (sources of forces) is the ***gauge potential***, $A_\mu(p)$, the field of vector potentials at each point. This field of connections describes how the associated section

(victims of forces)—the charged matter field or the Dirac wavefunction field—twists and turns as p moves.

Just as $\Gamma^\lambda_{\mu\nu}(p)$ corrects for the twisting and turning of tangent vectors at each point p , $A_\mu(p)$ corrects for the twisting of the *internal phase* of the charged matter field at each point p .

Just as gravitational forces arise from the Levi-Civita or spin connection, electrodynamical forces arise from the gauge potential. We now finalize our slogan:

Forces arise from connections on sections of principal bundles.

Invariance

If the Lagrangian density, $\mathcal{L}$, of gravitation is invariant with respect to smooth frame choice at each point, the physics is **locally Lorentz invariant**.

If the Lagrangian density, $\mathcal{L}$, of electrodynamical gauge theory is invariant with respect to smooth choice of external unitary phase from $U(1)$, the physics is **locally gauge invariant**.

Invariant here means numerically equal and of the same algebraic form, as explained in my earlier notebook, *From the steam age to quantum fields*.

We study Invariance at length in future notebooks.

Conclusion

In this work, we bridge the conceptual and computational gap between general relativity for gravitation and gauge theory for non-gravitational forces. By treating “gauge” not just as mathematics but as a concrete choice of measurement basis and origin—whether for position in spacetime or for phase in a complex field—we unify the origin of forces under a single slogan: **Forces arise from connections on sections of principal bundles**.

We showed that “garbage terms” generated by differentiating a phase-shifted field are mathematically analogous to “garbage terms” generated by differentiating a vector in curvilinear spacetime coordinates. In both cases, a connection field—the vector potential A_μ or the Christoffel Symbols $\Gamma^\mu_{\nu\lambda}$ —cancels these terms and restores phase covariance and geometrical covariance, respectively.

Computationally, the Relativity Toolkit as of v1.11.0 is “doubly covariant.” It automatically produces identities for hybrid objects like the Charged Vector Boson W^α , which “ride on” spacetime geometry while twisting in an internal gauge space. $\mathcal{UD}$ ’s symbolic machinery is ready for more of the Standard Model.

While this notebook focuses on the Abelian (commutative) $U(1)$ sector, the geometric machinery sets the stage for generalization to non-Abelian gauge theories incorporating $SU(2)$ and $SU(3)$.

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